



UNIVERSIDAD DE LAS FUERZAS ARMADAS - ESPE

DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN
Ingeniería de la Seguridad del Software

Integrantes: Mateo Sosa

NRC: 30808

Fecha: April 30, 2026

Contents

Objetivos de Aprendizaje	3
Topología del Experimento	3
Marco Teórico	3
Escaneo	4
Enumeración	4
Análisis de Vulnerabilidades	4
Desarrollo	5
Reconocimiento Inicial	5
Escaneo de Puertos y Versiones	6
Enumeración de Servicios Específicos	7
Script y Análisis de Vulnerabilidades	8
Ejercicios	10
Escaneo general de vulnerabilidades	10
Escaneo rápido masivo	11
<i>Fingerprinting</i> y detección de versiones	12
Uso de Categorías NSE	13
Escaneo de vulnerabilidades SSL/TLS	14
Escaneo centrado en SMB/Windows	15
Conclusiones	15

List of Figures

1	Revisión de Conectividad entre Metasploitable y Kali	5
2	Escaneo de Servicios Detallado	6
3	Extracción de Información	7
4	Script de Vulnerabilidades	8
5	Ejecución del Script de Vulnerabilidades sobre Metasploitable	9

6	Panorama Rápido de Servicios y Posibles Vulnerabilidades	10
7	Identificación de Hosts Activos y Puertos Abiertos	11
8	Identificación de Sistema Operativo y Versiones	12
9	Detección de Problemas en Servicios Web y Aplicaciones	13
10	Identificación de Problemas en Capa TLS/SSL	14
11	Enumeración de Servicios SMB y Posibles Vulnerabilidades Relacionadas	15

Objetivos de Aprendizaje

Al finalizar esta práctica, usted será capaz de:

- Comprender la diferencia fundamental entre el reconocimiento pasivo y activo, y la enumeración de servicios.
- Identificar los flags (`-sS`, `-sV`, `-O`, `-sU`) y las técnicas clave de Nmap para realizar un mapeo completo de un sistema objetivo.
- Aplicar herramientas de escaneo y enumeración (Nmap y Enum4linux) en un entorno controlado (laboratorio virtual).
- Analizar y documentar los hallazgos para determinar posibles vectores de ataque iniciales.
- Aplicar el script `nmap --script=vuln` para automatizar la identificación de vulnerabilidades conocidas (Common Vulnerabilities and Exposures, CVEs) y configuraciones débiles en los servicios activos.

Topología del Experimento

El entorno debe ser aislado (red interna o Host-Only) para garantizar que el tráfico del taller no salga al exterior, cumpliendo con la ética profesional.

Rol	Sistema Operativo (VM)	Propósito	Configuración de Red
Atacante (Scanner)	Kali Linux (vía VirtualBox/VMware)	Ejecución de herramientas de escaneo y enumeración (Nmap, Enum4linux)	Adaptador de Red Interna (ej. 192.168.56.10)
Objetivo A (Víctima)	Metasploitable 2 (Linux vulnerable)	Host con múltiples vulnerabilidades y puertos abiertos, ideal para el escaneo	Adaptador de Red Interna (ej. 192.168.56.11)
Objetivo B (Simulación)	Ubuntu Desktop (o Windows Server de prueba)	Simula un entorno SMB/Samba para practicar la enumeración	Adaptador de Red Interna (ej. 192.168.56.12)

Marco Teórico

La Fase 2: Escaneo y Enumeración sigue al Reconocimiento (Footprinting) y busca interactuar activamente con el objetivo para extraer información detallada.

Escaneo

Es la etapa donde se envían paquetes a la red objetivo para identificar:

1. **Hosts Activos:** Usando ICMP o barrido de ping.
2. **Puertos Abiertos:** Utilizando técnicas como SYN Scan (**-sS**), que es más sigiloso que un escaneo de conexión (**-sT**) porque no completa el handshake TCP.
3. **Servicios y Versiones:** Determinando qué software específico se ejecuta en cada puerto abierto (ej. Apache, OpenSSH).

Enumeración

Es el proceso de extracción intensiva de información de un servicio o sistema operativo específico. Se utiliza la información recopilada en la fase de escaneo para:

- Obtener nombres de usuario válidos y listas de grupos.
- Identificar recursos compartidos (SMB/Samba).
- Extraer políticas de seguridad a nivel de sistema.

Análisis de Vulnerabilidades

Utiliza los datos de la fase de escaneo y enumeración (servicios y versiones) para evaluar las debilidades de un sistema.

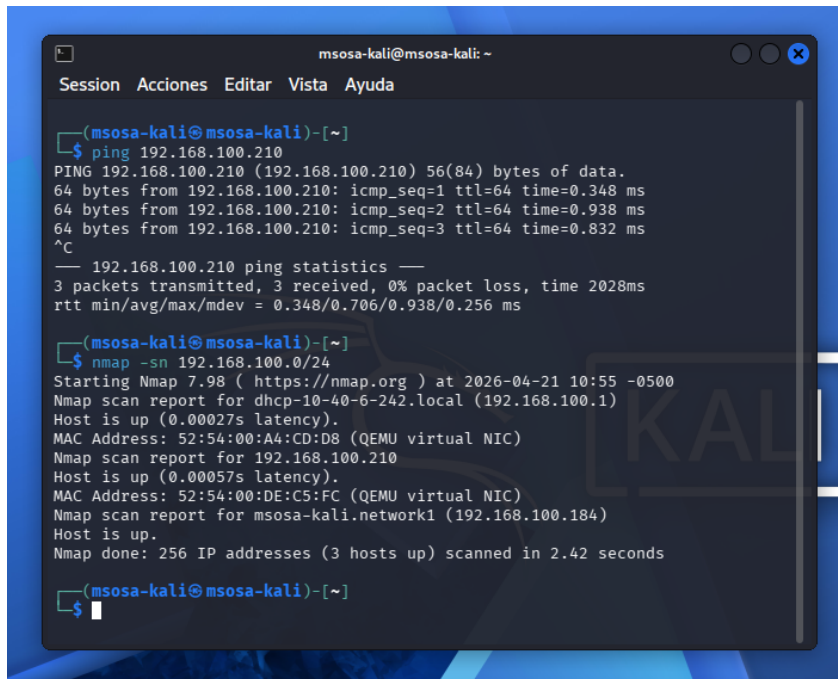
El script `nmap --script=vuln` es una herramienta esencial en esta fase. Permite:

- Detectar vulnerabilidades conocidas (CVEs).
- Identificar posibles backdoors.
- Verificar el uso de credenciales por defecto.
- Detectar debilidades en protocolos como SMB, FTP o SSH.

Al automatizar esta búsqueda, el ethical hacker puede priorizar objetivos para la siguiente fase: la explotación.

Desarrollo

Reconocimiento Inicial



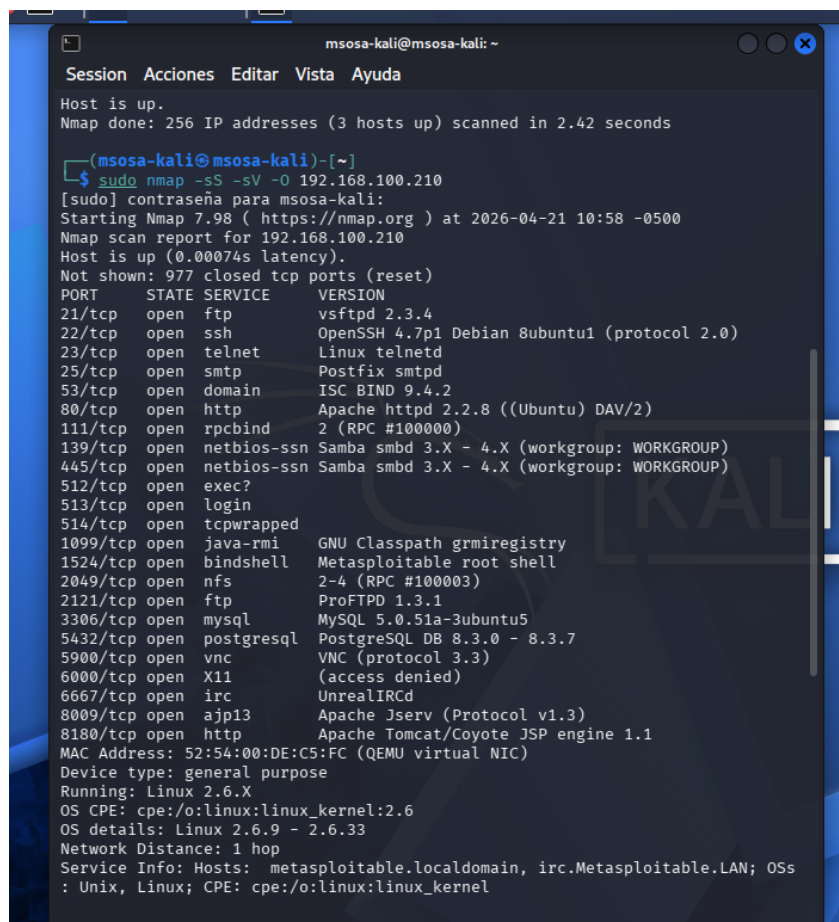
```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
  
(msosa-kali@msosa-kali)-[~]  
$ ping 192.168.100.210  
PING 192.168.100.210 (192.168.100.210) 56(84) bytes of data.  
64 bytes from 192.168.100.210: icmp_seq=1 ttl=64 time=0.348 ms  
64 bytes from 192.168.100.210: icmp_seq=2 ttl=64 time=0.938 ms  
64 bytes from 192.168.100.210: icmp_seq=3 ttl=64 time=0.832 ms  
^C  
--- 192.168.100.210 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2028ms  
rtt min/avg/max/mdev = 0.348/0.706/0.938/0.256 ms  
  
(msosa-kali@msosa-kali)-[~]  
$ nmap -sn 192.168.100.0/24  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 10:55 -0500  
Nmap scan report for dhcp-10-40-6-242.local (192.168.100.1)  
Host is up (0.00027s latency).  
MAC Address: 52:54:00:A4:CD:D8 (QEMU virtual NIC)  
Nmap scan report for 192.168.100.210  
Host is up (0.00057s latency).  
MAC Address: 52:54:00:DE:C5:FC (QEMU virtual NIC)  
Nmap scan report for msosa-kali.network1 (192.168.100.184)  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.42 seconds  
  
(msosa-kali@msosa-kali)-[~]  
$
```

Figure 1: Revisión de Conectividad entre Metasploitable y Kali

Para iniciar con el desarrollo, se debe constar una conexión entre ambos dispositivos, tal como se observa en la figura 1.

Se detectaron 3 dispositivos en la misma red.

Escaneo de Puertos y Versiones

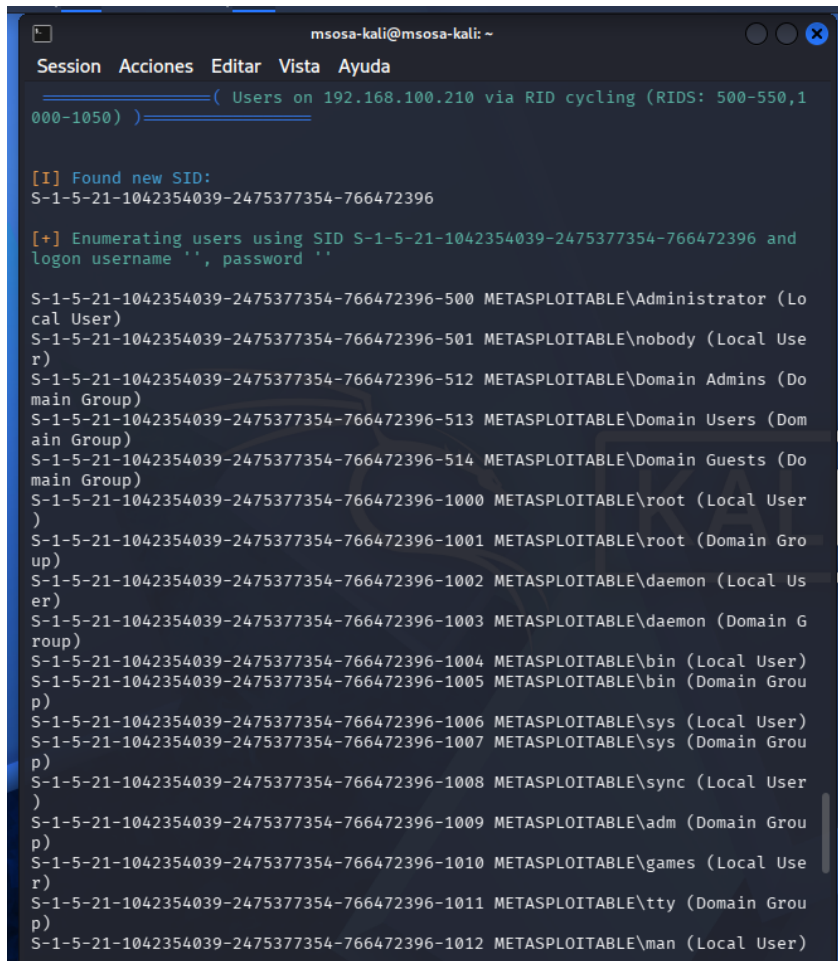


```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.42 seconds  
  
(msosa-kali@msosa-kali)-[~]  
$ sudo nmap -sS -sV -O 192.168.100.210  
[sudo] contraseña para msosa-kali:  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 10:58 -0500  
Nmap scan report for 192.168.100.210  
Host is up (0.00074s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec?          
513/tcp   open  login          
514/tcp   open  tcpwrapped     
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 52:54:00:DE:C5:FC (QEMU virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs : Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 2: Escaneo de Servicios Detallado

La figura 2 detecta que la máquina virtual de Metasploitable tiene una versión de Linux 2.6.X, cuyo kernel es *linux_kernel:2.6*.

Enumeración de Servicios Específicos



```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
===== ( Users on 192.168.100.210 via RID cycling (RIDS: 500-550,1  
000-1050) ) =====  
  
[I] Found new SID:  
S-1-5-21-1042354039-2475377354-766472396  
  
[+] Enumerating users using SID S-1-5-21-1042354039-2475377354-766472396 and  
logon username '', password ''  
  
S-1-5-21-1042354039-2475377354-766472396-500 METASPLOITABLE\Administrator (Lo  
cal User)  
S-1-5-21-1042354039-2475377354-766472396-501 METASPLOITABLE\nobody (Local Use  
r)  
S-1-5-21-1042354039-2475377354-766472396-512 METASPLOITABLE\Domain Admins (Do  
main Group)  
S-1-5-21-1042354039-2475377354-766472396-513 METASPLOITABLE\Domain Users (Dom  
ain Group)  
S-1-5-21-1042354039-2475377354-766472396-514 METASPLOITABLE\Domain Guests (Do  
main Group)  
S-1-5-21-1042354039-2475377354-766472396-1000 METASPLOITABLE\root (Local User  
)  
S-1-5-21-1042354039-2475377354-766472396-1001 METASPLOITABLE\root (Domain Gro  
up)  
S-1-5-21-1042354039-2475377354-766472396-1002 METASPLOITABLE\daemon (Local Us  
er)  
S-1-5-21-1042354039-2475377354-766472396-1003 METASPLOITABLE\daemon (Domain G  
roup)  
S-1-5-21-1042354039-2475377354-766472396-1004 METASPLOITABLE\bin (Local User)  
S-1-5-21-1042354039-2475377354-766472396-1005 METASPLOITABLE\bin (Domain Grou  
p)  
S-1-5-21-1042354039-2475377354-766472396-1006 METASPLOITABLE\sys (Local User)  
S-1-5-21-1042354039-2475377354-766472396-1007 METASPLOITABLE\sys (Domain Grou  
p)  
S-1-5-21-1042354039-2475377354-766472396-1008 METASPLOITABLE\sync (Local User  
)  
S-1-5-21-1042354039-2475377354-766472396-1009 METASPLOITABLE\adm (Domain Grou  
p)  
S-1-5-21-1042354039-2475377354-766472396-1010 METASPLOITABLE\games (Local Use  
r)  
S-1-5-21-1042354039-2475377354-766472396-1011 METASPLOITABLE\tty (Domain Grou  
p)  
S-1-5-21-1042354039-2475377354-766472396-1012 METASPLOITABLE\man (Local User)
```

Figure 3: Extracción de Información

Como se puede observar en la figura 3, se hizo un ataque de fuerza bruta al objetivo, aunque no pudo encontrar ningún usuario o dominio.

Script y Análisis de Vulnerabilidades



```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
GNU nano 8.7.1 script_vuln.sh  
IP_ADDRESS='192.168.56.210'  
  
nmap -F -sV -v $IP_ADDRESS  
  
sudo nmap -sV --script=vuln -v $IP_ADDRESS  
  
nmap -sV -v $IP_ADDRESS  
  
nmap -sV -p 1-65535 -T4 $IP_ADDRESS  
  
nmap -sV --script=banner $IP_ADDRESS  
  
[ 11 líneas leídas ]  
^G Ayuda      ^O Guardar    ^F Buscar     ^K Cortar     ^T Ejecutar  
^X Salir      ^R Leer fich. ^\ Reemplazar  ^U Pegar      ^J Justificar
```

Figure 4: Script de Vulnerabilidades

El script de vulnerabilidades es el siguiente:

```
# IP de Metasploitable  
IP_ADDRESS='192.168.56.210'  
  
# Comandos para vulnerar  
nmap -F -sV -v $IP_ADDRESS  
  
sudo nmap -sV --script=vuln -v $IP_ADDRESS  
  
nmap -sV -v $IP_ADDRESS  
  
nmap -sV -p 1-65535 -T4 $IP_ADDRESS  
  
nmap -sV --script=banner $IP_ADDRESS
```



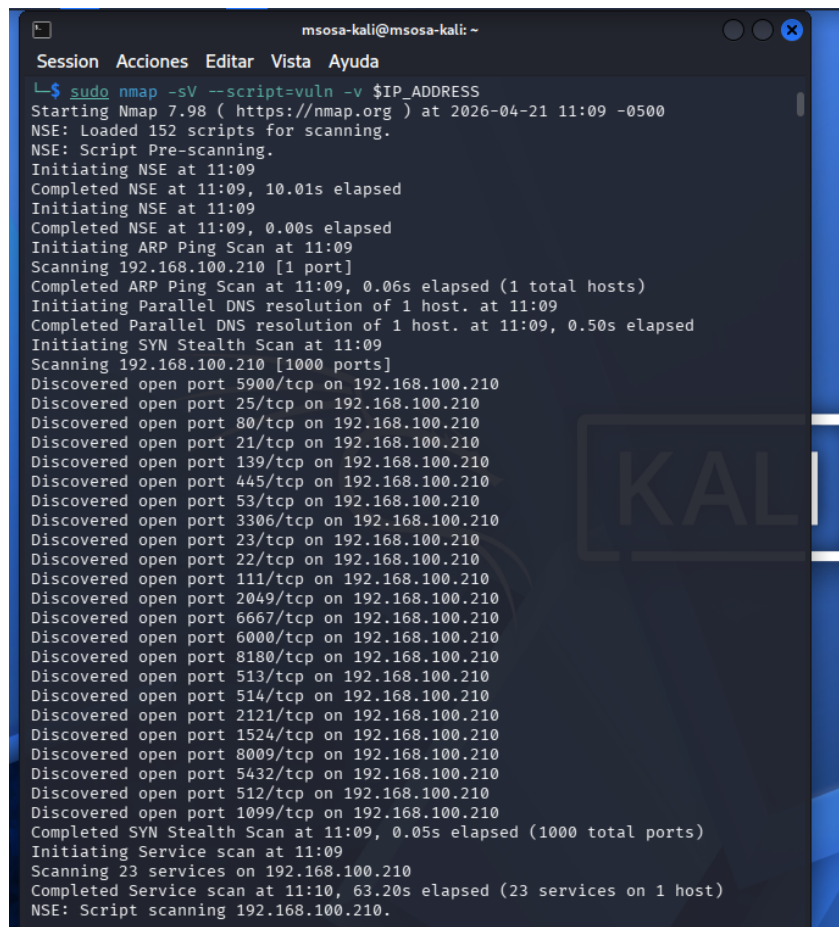
```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
(msosa-kali@msosa-kali)-[~]  
$ sh script_vuln.sh  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 11:06 -0500  
NSE: Loaded 48 scripts for scanning.  
Initiating Ping Scan at 11:06  
Scanning 192.168.56.210 [4 ports]  
Completed Ping Scan at 11:06, 3.02s elapsed (1 total hosts)  
Nmap scan report for 192.168.56.210 [host down]  
Read data files from: /usr/share/nmap  
Note: Host seems down. If it is really up, but blocking our ping probes, try  
-Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.17 seconds  
Raw packets sent: 8 (304B) | Rcvd: 0 (0B)  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 11:06 -0500  
NSE: Loaded 152 scripts for scanning.  
NSE: Script Pre-scanning.  
Initiating NSE at 11:06  
Completed NSE at 11:06, 10.01s elapsed  
Initiating NSE at 11:06  
Completed NSE at 11:06, 0.00s elapsed  
Initiating Ping Scan at 11:06  
Scanning 192.168.56.210 [4 ports]  
Completed Ping Scan at 11:06, 3.03s elapsed (1 total hosts)  
Nmap scan report for 192.168.56.210 [host down]  
NSE: Script Post-scanning.  
Initiating NSE at 11:06  
Completed NSE at 11:06, 0.00s elapsed  
Initiating NSE at 11:06  
Completed NSE at 11:06, 0.00s elapsed  
Read data files from: /usr/share/nmap  
Note: Host seems down. If it is really up, but blocking our ping probes, try  
-Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 13.32 seconds  
Raw packets sent: 8 (304B) | Rcvd: 0 (0B)  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 11:06 -0500  
NSE: Loaded 48 scripts for scanning.  
Initiating Ping Scan at 11:06  
Scanning 192.168.56.210 [4 ports]  
Completed Ping Scan at 11:07, 3.02s elapsed (1 total hosts)  
Nmap scan report for 192.168.56.210 [host down]  
Read data files from: /usr/share/nmap  
Note: Host seems down. If it is really up, but blocking our ping probes, try  
-Pn
```

Figure 5: Ejecución del Script de Vulnerabilidades sobre Metasploitable

En la ejecución del script mostrada en la figura 5, se aprecian los fallos de seguridad concretos de la máquina virtual.

Ejercicios

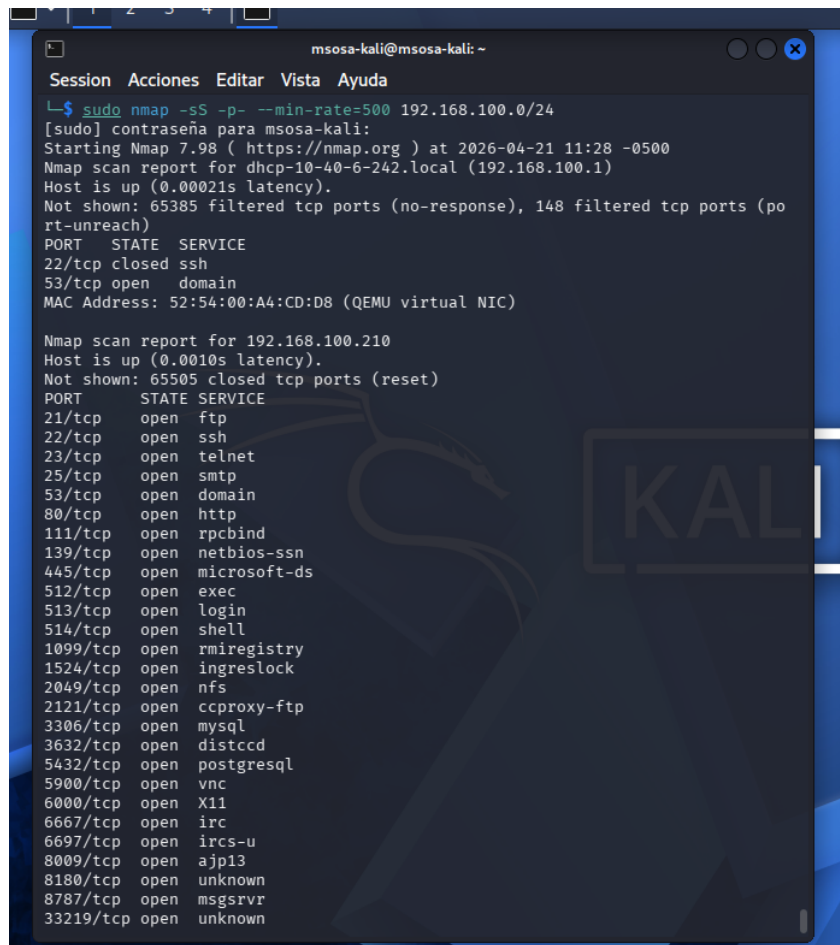
Escaneo general de vulnerabilidades



```
msosa-kali@msosa-kali: ~  
Session Acciones Editar Vista Ayuda  
└─$ sudo nmap -sV --script=vuln -v $IP_ADDRESS  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 11:09 -0500  
NSE: Loaded 152 scripts for scanning.  
NSE: Script Pre-scanning.  
Initiating NSE at 11:09  
Completed NSE at 11:09, 10.01s elapsed  
Initiating NSE at 11:09  
Completed NSE at 11:09, 0.00s elapsed  
Initiating ARP Ping Scan at 11:09  
Scanning 192.168.100.210 [1 port]  
Completed ARP Ping Scan at 11:09, 0.06s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 11:09  
Completed Parallel DNS resolution of 1 host. at 11:09, 0.50s elapsed  
Initiating SYN Stealth Scan at 11:09  
Scanning 192.168.100.210 [1000 ports]  
Discovered open port 5900/tcp on 192.168.100.210  
Discovered open port 25/tcp on 192.168.100.210  
Discovered open port 80/tcp on 192.168.100.210  
Discovered open port 21/tcp on 192.168.100.210  
Discovered open port 139/tcp on 192.168.100.210  
Discovered open port 445/tcp on 192.168.100.210  
Discovered open port 53/tcp on 192.168.100.210  
Discovered open port 3306/tcp on 192.168.100.210  
Discovered open port 23/tcp on 192.168.100.210  
Discovered open port 22/tcp on 192.168.100.210  
Discovered open port 111/tcp on 192.168.100.210  
Discovered open port 2049/tcp on 192.168.100.210  
Discovered open port 6667/tcp on 192.168.100.210  
Discovered open port 6000/tcp on 192.168.100.210  
Discovered open port 8180/tcp on 192.168.100.210  
Discovered open port 513/tcp on 192.168.100.210  
Discovered open port 514/tcp on 192.168.100.210  
Discovered open port 2121/tcp on 192.168.100.210  
Discovered open port 1524/tcp on 192.168.100.210  
Discovered open port 8009/tcp on 192.168.100.210  
Discovered open port 5432/tcp on 192.168.100.210  
Discovered open port 512/tcp on 192.168.100.210  
Discovered open port 1099/tcp on 192.168.100.210  
Completed SYN Stealth Scan at 11:09, 0.05s elapsed (1000 total ports)  
Initiating Service scan at 11:09  
Scanning 23 services on 192.168.100.210  
Completed Service scan at 11:10, 63.20s elapsed (23 services on 1 host)  
NSE: Script scanning 192.168.100.210.
```

Figure 6: Panorama Rápido de Servicios y Posibles Vulnerabilidades

Escaneo rápido masivo



The screenshot shows a terminal window titled 'msosa-kali@msosa-kali: ~'. The terminal displays the output of an Nmap scan command: `sudo nmap -ss -p- --min-rate=500 192.168.100.0/24`. The output shows the scan starting at 2026-04-21 11:28 -0500. The first host scanned is 192.168.100.1 (dhcp-10-40-6-242.local), which is up with a latency of 0.00021s. It has 148 filtered TCP ports. The second host scanned is 192.168.100.210, which is also up with a latency of 0.0010s. It has 65505 closed TCP ports (reset). The terminal also displays a list of open ports and services for the second host.

```
Session Acciones Editar Vista Ayuda
└─$ sudo nmap -ss -p- --min-rate=500 192.168.100.0/24
[sudo] contraseña para msosa-kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 11:28 -0500
Nmap scan report for dhcp-10-40-6-242.local (192.168.100.1)
Host is up (0.00021s latency).
Not shown: 65385 filtered tcp ports (no-response), 148 filtered tcp ports (po
rt-unreach)
PORT      STATE  SERVICE
22/tcp    closed ssh
53/tcp    open   domain
MAC Address: 52:54:00:A4:CD:D8 (QEMU virtual NIC)

Nmap scan report for 192.168.100.210
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE  SERVICE
21/tcp    open   ftp
22/tcp    open   ssh
23/tcp    open   telnet
25/tcp    open   smtp
53/tcp    open   domain
80/tcp    open   http
111/tcp   open   rpcbind
139/tcp   open   netbios-ssn
445/tcp   open   microsoft-ds
512/tcp   open   exec
513/tcp   open   login
514/tcp   open   shell
1099/tcp  open   rmiregistry
1524/tcp  open   ingreslock
2049/tcp  open   nfs
2121/tcp  open   ccproxy-ftp
3306/tcp  open   mysql
3632/tcp  open   distccd
5432/tcp  open   postgresql
5900/tcp  open   vnc
6000/tcp  open   X11
6667/tcp  open   irc
6697/tcp  open   ircs-u
8009/tcp  open   ajp13
8180/tcp  open   unknown
8787/tcp  open   msgsrvr
33219/tcp open   unknown
```

Figure 7: Identificación de Hosts Activos y Puertos Abiertos

Fingerprinting y detección de versiones

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-28 10:46 -0500
Nmap scan report for 192.168.100.210
Host is up (0.00065s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.100.184
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName
=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRF
Y, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-date: 2026-04-28T15:47:22+00:00; 0s from scanner time.
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2_RC4_128_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
```

Figure 8: Identificación de Sistema Operativo y Versiones

Uso de Categorías NSE

```
l-$ nmap -p 80,443 --script "http-enum,http-vuln*" -v 192.168.100.210
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-28 10:48 -0500
NSE: Loaded 25 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 10:48
Completed NSE at 10:48, 0.00s elapsed
Initiating ARP Ping Scan at 10:48
Scanning 192.168.100.210 [1 port]
Completed ARP Ping Scan at 10:48, 0.06s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 10:48
Completed Parallel DNS resolution of 1 host. at 10:48, 0.50s elapsed
Initiating SYN Stealth Scan at 10:48
Scanning 192.168.100.210 [2 ports]
Discovered open port 80/tcp on 192.168.100.210
Completed SYN Stealth Scan at 10:48, 0.01s elapsed (2 total ports)
NSE: Script scanning 192.168.100.210.
Initiating NSE at 10:48
Completed NSE at 10:48, 1.32s elapsed
Nmap scan report for 192.168.100.210
Host is up (0.00081s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
| http-enum:
|   /tikiwiki/: Tikiwiki
|   /test/: Test page
|   /phpinfo.php: Possible information file
|   /phpMyAdmin/: phpMyAdmin
|   /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'
|   /icons/: Potentially interesting folder w/ directory listing
|_  /index/: Potentially interesting folder
443/tcp    closed https
MAC Address: 52:54:00:DE:C5:FC (QEMU virtual NIC)

NSE: Script Post-scanning.
Initiating NSE at 10:48
Completed NSE at 10:48, 0.00s elapsed
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 2.11 seconds
Raw packets sent: 3 (116B) | Rcvd: 3 (112B)
```

Figure 9: Detección de Problemas en Servicios Web y Aplicaciones

Escaneo de vulnerabilidades SSL/TLS

```
(msosa-kali@msosa-kali)-[~]
$ nmap --script ssl-enum-ciphers -p 443 192.168.100.210
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-28 10:48 -0500
Nmap scan report for 192.168.100.210
Host is up (0.00089s latency).

PORT      STATE SERVICE
443/tcp   closed https
MAC Address: 52:54:00:DE:C5:FC (QEMU virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds

(msosa-kali@msosa-kali)-[~]
$
```

Figure 10: Identificación de Problemas en Capa TLS/SSL

Escaneo centrado en SMB/Windows

```
└─$ nmap -p 139,445 --script "smb-enum-users,smb-enum-shares,smb-vuln*" 192
.168.100.210
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-28 10:49 -0500
Nmap scan report for 192.168.100.210
Host is up (0.00060s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 52:54:00:DE:C5:FC (QEMU virtual NIC)

Host script results:
| smb-enum-users:
| METASPLOITABLE\backup (RID: 1068)
|   Full name: backup
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\bin (RID: 1004)
|   Full name: bin
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\bind (RID: 1210)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\daemon (RID: 1002)
|   Full name: daemon
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\dhcp (RID: 1202)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\distccd (RID: 1222)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\ftp (RID: 1214)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\games (RID: 1010)
|   Full name: games
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\gnats (RID: 1082)
|   Full name: Gnats Bug-Reporting System (admin)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\irc (RID: 1078)
|   Full name: ircd
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\klog (RID: 1206)
|   Flags: Account disabled, Normal user account
| METASPLOITABLE\libuuid (RID: 1200)
|   Flags: Account disabled, Normal user account
```

Figure 11: Enumeración de Servicios SMB y Posibles Vulnerabilidades Relacionadas

Conclusiones

- La fase de escaneo representa la transición de reconocimiento pasivo a activo.
- La combinación `nmap -sS -sV -O` es un estándar para obtener información detallada del objetivo.
- La enumeración transforma datos generales (puertos abiertos) en información útil para explotación (usuarios, recursos).
- Todas las prácticas deben ejecutarse en entornos aislados, respetando principios éticos del hacking.